Muhammad Saad Rafiq

Computer Science Professional| Web Developer | Software Developer

Haripur, Pakistan | saadrafiq033@gmail.com | +92 331 5143525

PROFESSIONAL SUMMARY

Results-driven Computer Science graduate with hands-on experience across the full software development lifecycle from frontend and backend web development to database design and mobile app development. Developed an Intelligent Surveillance System for Enhanced Safety at Construction Sites, a full-stack solution combining computer vision-based hazard detection with secure, role-based backend architecture, including authentication, JWT-based session management, and access control for different user roles (admin, supervisor, on-site personnel). Proficient in SQL, REST API design, and building full-stack web applications, with additional experience in mobile app development and applying machine learning and computer vision techniques to solve real-world safety and monitoring challenges. Strong foundation in data structures, algorithms, and object-oriented programming, with working knowledge of Python, Scikit-learn, and TensorFlow. Seeking an entry-level Software Engineer, Full-Stack Developer, or Data Analyst role where strong problem-solving skills and practical, project-driven experience can deliver measurable outcomes.

TECHNICAL SKILLS

Programming Languages: Python, C++, PHP, JavaScript

AI / ML Frameworks: YOLO (Ultralytics), DeepFace, OpenCV, TensorFlow, PyTorch

Computer Vision Techniques: Real-Time Object Detection, Face Recognition, Face Embedding & Matching, PPE Violation Detection, Video Frame Processing, Image Preprocessing

Backend Development: Node.js, Express.js, Flask, REST API Development, JWT Authentication, Role-Based Access Control, bcrypt, Multer (File Uploads), node-cron (Task Scheduling)

Database: MySQL, Relational Schema Design, Complex SQL Queries

Web Technologies: HTML5, CSS3, JavaScript (ES6+), REST APIs, WebRTC / Media Devices API, Web Audio API, Responsive UI Design

Video Streaming & Networking: RTSP, FFmpeg, MJPEG Streaming, Multithreading, Socket-Based Frame Piping

Tools & Platforms: VS Code, Antigravity IDE, Jupyter Lab, Google Colab, Git, GitHub, Windows/Linux, Flask

Core ML Competencies: Supervised & Unsupervised Learning, Neural Networks, Classification, Regression, Data Preprocessing, Model Evaluation, OOP, SMOTE

WORK EXPERIENCE

Data Entry Specialist

April 2021 – August 2022

Sheraz Minerals

- Managed records of incoming raw materials, including quantities, dates, and vehicle details
- Maintained grinding and production data, such as powder quantities and percentages
- Recorded sales information, including customers, quantities, dates, and material percentages
- Ensured all inventory, production, and sales records were accurate, organized, and up to date

INTERNSHIP EXPERIENCE

Web Developer Intern

Jul 2023 – Sep 2023

MTech

- Designed and developed multiple responsive front-end web pages using HTML, CSS, and JavaScript, ensuring consistent layout and functionality across devices.
- Worked on back-end development using PHP integrated with a MySQL database to support dynamic content and data-driven pages.
- Collaborated on improving website structure and user experience, contributing to a more functional and visually consistent company website.

PROJECTS

Intelligent Surveillance System for Enhanced Safety at Construction Sites [Final Year Project]

Tech Stack: Node.js · Express.js · Python · Flask · YOLO · DeepFace · OpenCV · MySQL · JavaScript · RTSP/FFmpeg

- Engineered an end-to-end intelligent construction-site safety platform integrating real-time CCTV monitoring, YOLO-based PPE violation detection, and DeepFace facial recognition for automated worker identification and attendance logging.
- Designed a microservice-style architecture connecting a Node.js/Express REST API with independent Python/Flask AI services for face authentication and live PPE detection, communicating via RESTful webhooks for instant violation reporting.

- Built an RTSP/FFmpeg-based video streaming pipeline with multithreaded frame processing, enabling low-latency MJPEG delivery of AI-annotated camera feeds to the web dashboard.
- Automated the full payroll lifecycle by linking attendance, violations, and fines to a dynamic salary engine with cron-based daily absence checks, auto-checkout, and real-time recalculation.
- Implemented role-based JWT authentication across 7 user roles (admin, supervisor, HR, monitor, etc.) with secured, permission-gated access to dashboard modules.
- Delivered a complete production-style system spanning database design (MySQL), backend APIs, AI inference services, and a responsive multi-page frontend dashboard for attendance, violations, health alerts, and payroll management.

North Trails Tourism Website
Tech Stack: HTML · CSS · JavaScript · PHP · MySQL

- Developed a full-stack tourism website showcasing hotels, tourist spots, and travel information, with a MySQL database to store and manage listings and booking-related data.
- Designed dynamic web pages to display hotel details, tourist attractions, and location-based information, with PHP handling backend logic and database interactions.
- Implemented core functionality including adding, searching, and displaying hotel and tourist spot records, focusing on a clean, user-friendly interface for browsing travel content.

Mobile Store Management System
Tech Stack: Java · Object-Oriented Programming (OOP) · GUI Development

- Built a GUI-based Mobile Store Management System in Java applying core OOP principles such as encapsulation, inheritance, and polymorphism to model products, sales, and inventory.
- Designed an interactive user interface for managing mobile phone inventory, including adding, updating, and searching product records in real time.
- Implemented functionality for tracking stock levels and processing sales transactions, using in-memory data structures to manage records during runtime, with a focus on clean class design and modular code.

Blood Bank Management System
Tech Stack: C# · MySQL · Windows Forms · ADO.NET · SQL

- Built a GUI-based Blood Bank Management System in C#, integrating a MySQL database to store and manage donor, recipient, and blood inventory records.
- Designed an interactive Windows Forms interface for registering donors, tracking blood group availability, and managing requests, with data persisted using ADO.NET for database connectivity.
- Implemented core functionality including adding, searching, and updating donor and inventory records, focusing on clean form design and a user-friendly interface.

EDUCATION

Bachelor of Science in Computer Science 2022 – 2026
Comsats University Islamabad, Abbottabad Campus.

Intermediate – Computer Science
The Punjab School & College. 2020 – 2021
The Peace College. 2019 – 2020

ADDITIONAL INFORMATION

Languages: English (Professional Proficiency) | Urdu (Native) | Hindko (Conversational)
Soft Skills: Cross-functional Collaboration · Analytical Thinking · Rapid Adaptability · Written & Verbal Communication · Problem Solving · Attention to Detail